

Satyabrata Bharati

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Professional Summary

Over 15 years of expertise in software architecture, design, and development across Automotive, Telecom, and Aerospace domains, specializing in **C++ (11/14/17)**, Linux and QNX for real-time embedded systems.

Expert in C++, Linux, and QNX, with hands-on experience in in modern C++, object-oriented programming, STL, multithreading, and memory management.

Experienced in architecture definition, system integration, and cross-functional team leadership.

Proficient in Agile methodologies and DevOps tools, ensuring efficient software lifecycle management.

Core Competencies

Software Architecture & System Design: Designing scalable, robust systems for real time applications.

C++ Programming: Expertise in **C++ 11/14/17**, STL, templates, design patterns, memory management, Multithreading, and object-oriented programming.

System Programming: Proficient in IPC (Shared Memory, POSIX Message Queues), TCP/IP socket programming, and low-level system optimization.

Embedded Systems: Real-time OS (Linux, QNX) and hardware interfacing (RCAR-S4, Raspberry Pi).

Adaptive AUTOSAR, functional safety: Middleware, functional safety (ISO 26262, ASIL), platform health monitoring and Time Synchronization (PTP, qgPTP, TSYNC) for automotive ECUs.

DevOps & CI/CD: Automation with Kubernetes, Docker, GitHub, and TeamCity.

Debugging & Optimization: Tools like GDB, Valgrind, perf, and gperf for performance tuning.

Technical Skills

Languages: C++ (11/14/17), Python, Shell Scripting

Operating Systems: Linux, QNX

System Programming: IPC, TCP/IP sockets, POSIX, multithreading, memory management

Tools & Frameworks: Google Protocol Buffers, ZMQ, Pulsar, Boost

DevOps & CI/CD: Kubernetes, Docker, GitHub, TeamCity, GitLab, JIRA

Development Tools & Monitoring: CMake, Conan, GTest, GCOV, Testwell CTC++, Parasoft, GitLab, MISRA C++, Nexus, JFROG, Prometheus, Grafana

Certifications

Functional Safety Engineer Level 1 (ISO 26262:2018)

Work Experience

KPIT Technologies | Architect (Apr 2023 – Present)

- Architected and led the Platform Health Monitoring (PHM) system and Time Synchronization TSYN module ensuring robustness in safety-critical environments for AUTOSAR-compliant middleware stacks for automotive ECUs.
- Developed high-precision time synchronization mechanisms using PTP, qgPTP, and hardware-assisted timestamping.
- Designed fail-safe and fault-tolerant mechanisms for automotive embedded platforms.
- Ensured compliance with ISO 26262 functional safety standards.

Tools: C++ 11/14, QNX, Conan, POSIX, IPC (Shared Memory, Message Queue), Conan build, Cmake Build System, GDB, JIRA, GitLab, Parasoft, Gtest, Gitlab, CI/CD Automation Build

Persistent Systems | Architect (Dec 2020 – Apr 2023)

- Led the design and development of service assurance solutions for IP-based communication networks.
- Developed applications for real-time network monitoring, fault detection, and system diagnostics.

Tools: C++ 11/14/17, Python, Bash, ZMQ, IPC, DevOps tools (Kubernetes, Docker, Terraform, Prometheus, Grafana).

Mphasis Ltd | Project Lead (May 2016 – Dec 2020)

- Designed and developed C++ applications for telecom and enterprise platforms.
- Led teams in software migration, performance tuning, and system integration.
- Developed applications with C++, multithreading, IPC, and database integrations.

Vodafone Shared Services | Deputy Manager (Oct 2014 – May 2016)

- Managed telecom billing systems and sales commission applications using C++, SQL/PLSQL, UNIX.
- Led a team in troubleshooting, feature enhancements, and system stability improvements.

At Mphasis | Module Lead Jul (2007 - Oct 2014)

PROVIS – Vodafone, Germany.

PROVIS is a commission calculation system at Vodafone Germany. The system is responsible for loading/maintaining transactions, master/reference data, calculating monthly commission, and generating statements/bills/reports/interfaces. The system has interfaces with other telecommunication systems at Vodafone such as KIAS2000, Voras, VITA, DWH, FHO and SAP.

At CMG Pvt Ltd | IT Consultant (Sep 2006 - June 2007)

PSA – Pre-Delivery Service Agent

PSA is a rating system that does real-time rating for all the data calls (SMS, MMS and GPRS). The system is implemented with most of the popular mobile service providers (Vodafone, STC, T-Mobile, SingTel). This system can communicate with third party Billing system (Nortel, Ericsson, Alcatel, etc.,) using specific interfaces.

C++, SQL, UNIX

At HCL TECHNOLOGIES | Member Technical Staff (Oct 2004 - Aug 2006)

NetBacker – is a product developed to backup, restore and replicate data across the IP network. It is based on Network Data Management Protocol (NDMP), which defines a mechanism for controlling transfer of data between primary and secondary storage using a client-server model. NetBacker comprises of Data Management Application (DMA), Data Service Provider (DSP) and Tape Service Provider (TSP). The communication between NDMP servers and the DMA takes place over bi-directional TCP/IP connections via XDR streams.

C, UNIX internals (IPC, sockets), RPC, NDMP, SCSI, UNIX

Education

B.E. (Information Technology) | Biju Patnaik Technical University, Odisha (2004, 70.07%)

Personal Details

Languages: English, Hindi, Oriya